

ChatGPT Mobile Voice feature India

This proposed solution aims to increase the number of voice searches made using ChatGPT's mobile voice Feature in India by addressing the Accuracy of voice interpretation.

Contributors: Sahil Bose

Status: Brainstorming/In Review/In Development

Launching on:

Resources: Link to previous PRDs etc

Problem Definition

- **What is the problem?** Users find Chatgpt voice feature inaccurately interprets them.
- **Who is facing the problem?** Users aged 23-44, tier 1, mobile first, tech savvy, working professionals.
- **What is the business value that will be unlocked by solving the problem?** Increased Retention, Higher Revenue, Market expansion
- **How will the target users benefit if the problem is solved?** Increased Hands free productivity, Users will multitask efficiently.
- **Why is it urgent to solve this problem now?** Voice adoption in India is growing at a high rate, Reduce Churn Rate

Goals

1. Increase successful voice interactions by improving speech recognition accuracy for Indians.
2. Increase user trust and satisfaction with the Voice interactions.
3. Increase long-term retention of voice users.

Metrics

- **Functional**
 1. Number of Voice Searches
 2. Speech Recognition Accuracy
 3. Voice Prompt Correction Rate (Reduce)
 4. Voice Profile Completion Rate. (Subject to Use of the Proposed Feature)
- **Non-Functional Metrics**
 1. Voice profile creation time < 40s
 2. Voice recognition latency<100ms (Should not increase)
 3. Voice Embedding generation success >99%

Non-Goals

The scope of this project intentionally excludes:

1. Improving ChatGPT response quality or reasoning accuracy.
2. Redesigning the overall voice interface.
3. Solving network-related latency issues.
4. Supporting desktop voice interactions.
5. Enhancing text-based ChatGPT interactions.

Validation of the problem

User Survey and Interviews were done for 20+ users which resulted in the following data:

- 65% of the Users find Typing faster/less time consuming.
- 30% users report wrong interpretation while using voice while other set of users (30% of the total) report going through multiple amendments of the transcription.
- Statements such as “Repeating myself makes me frustrated, I would rather type right the first time than waste time repeating myself.” Were reported during interviews.

Understanding the target audience

- **User Segment in Focus:** Age 23-44, Tier 1 City , Tech Savvy
- **Segment Size:** 4.8 Million Current Users + 4.8 New Users
- [User Persona](#)

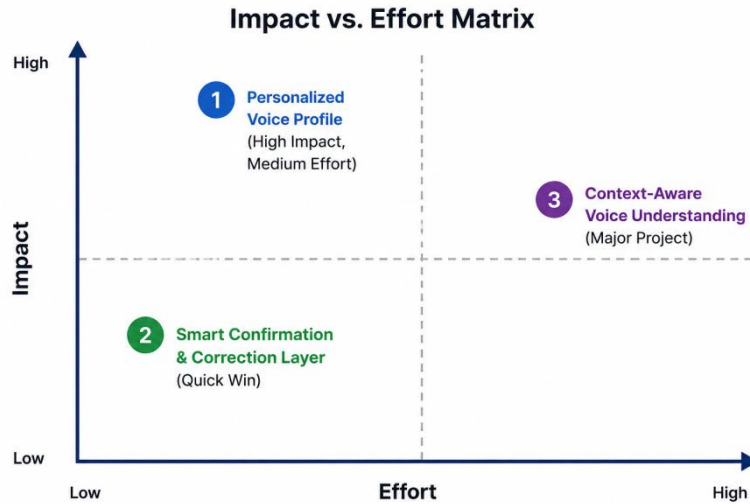
Solution

Proposed Solutions and Comparison

Solution	Description
1. Personalized Voice Profile (Accent Learning)	Users complete a short 30–60 second voice calibration that learns their accent, pronunciation, and speaking style. The voice profile continuously adapts based on future corrections, improving speech recognition accuracy over time.
2. Smart Confirmation & Correction Layer	When ChatGPT detects low-confidence words in a voice prompt, it highlights them before submission, allowing users to quickly confirm or edit them. Corrections are used to improve future recognition.
3. Context-Aware Voice Understanding	ChatGPT leverages conversation history, user preferences, and current task context to intelligently infer ambiguous words and phrases, reducing errors without requiring additional user input.

The above Proposed solutions were compared on ICE and Impact/Effort matrix and the following were the findings.

Solution	Impact	Confidence	Ease	ICE Score
Personalized Voice Profile	10	7	5	350
Smart Confirmation Layer	6	6	9	324
Context-Aware Voice Understanding	8	7	4	224



Personalized Voice Profile was chosen as the solution to increase accuracy of voice interpretation Based on the ICE score and Impact/Effort matrix . Instead of simply reducing errors, it creates a highly personalized experience by adapting to each user's speech patterns over time. This directly addresses one of the biggest pain points uncovered in user research.

Key Insight: Users already understand personalization from Face ID and fingerprint unlocking. A "Voice Profile" feels intuitive.

- [User Flow](#)
- [Wireflow](#)
- **Key Features:** Personalized Voice Profile, a 40 seconds onboarding after repeated recognition failures.
- **Key Logic:**
 - Trigger Voice Profile recommendation after repeated correction events.
 - Store voice embeddings instead of raw audio.
 - Use embeddings during speech decoding.
 - Continuously update embeddings using confirmed corrections.

- **Additional Data Types**

Field	Type	Purpose
user_id	UUID	Links the profile to the user
embedding_vector	Vector	Numerical representation of the user's voice characteristics
language_preferences	Array	Languages frequently used (e.g., English, Hindi, Telugu)
last_updated	Timestamp	Tracks when the profile was last refined
profile_status	Enum	Active / Training / Disabled
confidence_score	Float	Indicates confidence in the learned profile
training_version	Integer	Supports future model upgrades

Launch Readiness

- **Key Milestones**

Milestone	Timeline
UX & Prototype Complete	Week 0-2
Engineering Development	Weeks 3–6
Internal QA	Weeks 7
Dogfooding (Employees)	Week 8
Beta Rollout (5% Users)	Week 9
Full Launch	Week 12

- **Internal Stakeholders**

1. Product Managers
2. Design (UX/UI)
3. Speech AI / ML Team
4. Mobile Engineering (Android & iOS)
5. Backend Engineering
6. Data Science
7. Privacy & Legal
8. QA
9. Customer Support
10. Analytics Team

- **Launch Checklist:**

- 1. Product**

- Success metrics
- Analytics dashboard ready

- 2. Engineering**

- Voice profile prompt (Frontend)
- Voice embedding storage(backend)
- Rollback Plan tested
- Monitoring dashboards configured

- 3. Design**

- Accessibility review completed
- User consent and privacy messaging approved
- Voice Profile onboarding reviewed
- Empty states, error states, and retraining prompts reviewed

- 4. Privacy & Legal**

- User Consent added
- Voice embedding Storage policy
- Delete profile flow verified

- 5. Customer Support**

- FAQ prepared
- Help center article published

- Help Escalation documentation prepared
- **Experimentation plan :**
 1. **A/B Test:** Users Aged 23-44, 2-4 weeks, **Control:** Existing Voice Experience, **Variant:** Voice Profile onboarding

Open Questions & Decisions Taken

- **Decisions Taken**
 1. Personalization chosen over generic model improvements.
 2. Voice embeddings used instead of raw audio.
 3. Voice Embedding over Raw Voice ? **Lower storage (mathematical representation of speech), Faster recognition, better privacy, Reduced Compliance issues.**
 4. Voice Profile is optional.
 5. Retraining Question triggered after repeated corrections.
- **Open Questions**
 1. Should Voice Profile be mandatory? **No/Optional.**
 2. How many training sentences are needed? **Start with 7, Validation Required**
 3. How many Transcription edits before retraining required? **5/week to start with**
 4. Should multiple Voice Profiles be supported? **Out of Scope**
 5. Should retraining be automatic? **No, user consent required**
 6. How long should embeddings be retained? **Privacy review required**
- **Out of Scope (descoped)**
 1. Desktop/Web implementation
 2. Multiple user profile use
 3. Language detection Improvement
- **Trade-offs**
 1. Optional onboarding: Some users might skip the feature reducing impact.
 2. Personalized Voice Profile: Additional engineering effort and onboarding effort.
 3. Store voice embeddings instead of recordings: Limits future model retraining using raw audio.
 4. Mobile-first launch: Desktop/Web users do not benefit initially.